MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

1/ TYPES 7580W AND 8930

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal (see note 9)
See figure 1
Mounting position: Any
Weight: 4 ounces (113.4 grams) nominal (7580W)
6 ounces (170.1 grams) nominal (8930)

ABSOLUTE RATINGS:

Parameter: Unit:	Fl MHz	Ef Vac (Note 1)	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ehk Vdc	Ib mAdc	Pg1 W	Pg2 W
Maximum:									
Class C Telep:	500	6.0±10%	1,500	-250	300	±150	200	2	12
Class C Teleg:	500	6.0±10%	2,000	-250	300	±150	250	2	12
Class AB1:	500	6.0±10%	2,000	-250	500	±150	250	2	12
Class AB1 (8930):	500	6.0±10%	2,400	-250	500	±150	250	2	12
TEST CONDITIONS:	---	6.0	1,000	Adj	300	0	150	---	---

ABSOLUTE RATINGS:

Parameter: Unit:	Pp W	Pi W	T(seal) °C	(anode T core) °C	tk sec (min)	Cooling Note 2	Barometric pressure, reduced mmHg
Maximum:							
Class C Telep:	165	300	250	250	30	---	349
Class C Teleg:	250	500	250	250	30	---	349
Class AB1:	250	500	250	250	30	---	349
Class AB1 (8930):	350	500	250	250	30	---	349
TEST CONDITIONS:	---	---	---	---	120	Note 3	---

GENERAL:

Qualification - Required

1/ Type 8930 replaces tube types:
EIMAC 8930 and X651Z
Amperex DX393
ITT 504809-1

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1256	Electrode current (screen)	Both		0.65	II	Ic2	-7.0	+3.0	mAdc
1261	Electrode voltage (grid)	Both	Eb = 2,000 Vdc; Ec2 = 400 Vdc; Ec1/Ib = 67 mAdc	0.65	II	Ec1	-70	-100	Vdc
1266	Total grid current	7580W	Eb = 2,000 Vdc; Ib = 125 mAdc (see note 14)	0.65	II	Ic1	---	-15	µAdc
		8930	Eb = 2,500 Vdc; Ec1/Ib = 140 mAdc (see note 14)	0.65	II	Ic1	---	-15	µAdc
1266	Primary-grid emission (control)	Both	Ic1 = 70 mAdc; t = 120; anode and screen-grid floating	0.65	II	Isg1	---	-250	µAdc
1266	Primary-grid emission (screen)	Both	Ec1 = 0; t = 120; Ic2 = 100 mAdc; anode floating	0.65	II	Isg2	---	-250	µAdc
1301	Heater current	Both		0.65	II	If	2.30	2.90	Aac
2212	Pulse emission	Both	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; prr = 11 ±1; tp = 4,500 µs (min); Ef = 5.4 Vac; tr = tf = 25 µs; slope = 0.5 percent; ripple = 0.1 percent	0.65	II	Δik	---	200	ma
1372	Current division (long pulse method A)	Both	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ib = 1.0 a; prr = 11 ±1; tp = 4,500 µs (min)	0.65 0.65 0.65	II II II	egk ic1 ic2	2.0 --- ---	9.0 150 260	v ma ma
---	Interelement leakage (1)	8930	Grid-cathode; C = 0.25 µF; E = ±400 Vdc (see note 8)	0.65 0.65	II II	I I	--- ---	+1.5 -1.5	µAdc µAdc
---	Interelement leakage (2)	8930	Grid-screen; C = 0.25 µF; E = ±1,000 Vdc (see note 8)	0.65 0.65	II II	I I	--- ---	+1.5 -1.5	µAdc µAdc
---	Interelement leakage (3)	8930	Anode-screen; C = 0.001 µF; E = 8.0 kVdc (see note 8)	0.65	II	I	---	5.5	µAdc
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	Both		---	---	Cgp Cin Cout	--- 16.0 4.2	0.06 18.5 5.2	pF pF pF
1336	Heater-cathode leakage	Both	Ehk = +150 Vdc Ehk = -150 Vdc (see note 10)	--- ---	--- ---	Ihk Ihk	--- ---	+150 -150	µAdc µAdc

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
2214	RF useful output power	7580W	Class C amplifier; F = 470 to 500 MHz; Eb = 2,000 Vdc; Ec1 = -90 Vdc; Ec2 = 250 to 300 Vdc; Ic1 = 25 mAdc (max); Eg1/Ib = 250 mAdc; Ef = 5.5 Vac (see note 11)	---	---	Po	225	---	W
	Quality conformance inspection, part 3								
---	Life test (2)	7580W	Group C; AB1 amplifier; Eb = 2,000 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 75 mAdc; Eg1/Ib = 250 mAdc; RL = 4,000 ±100 ohms; anode tank Q = 10 to 15; F = 2 MHz (min), 10 MHz (max); t = 500 hours (see note 7)	---	---	Po (initial)	250	---	W(useful)
---	Life-test (2) end points:		See note 7	---	---	3rd IM	-22	---	dB
2204	Linear amplifier power output and distortion					5th IM	-27	---	dB
---	Life test (2)	7580W	See note 7	---	---	ΔIb	---	40	mAdc
						Po	225	---	W(useful)
---	Life test (1)	7580W	RF useful output power; t = 500 hours (see note 4)	---	---	---	---	---	---
---	Life-test (1) end points:		Ef = 6.0 Vac	---	---	Δik	---	100	ma
2212	Pulse emission		Ehk = +150 Vdc	---	---	Ihk	---	150	μAdc
1336	Heater-cathode leakage		Ehk = -150 Vdc	---	---	Ihk	---	150	μAdc
1266	Primary grid emission (control)			---	---	Isg1	---	-250	μAdc
1266	Primary grid emission (screen)			---	---	Isg2	---	-250	μAdc
---	Cooling	7580W	Ec1/Ib = 250 mAdc (see notes 4 and 5)	---	---	T (anode core)	---	250	°C
		8930	Eb = 1,400 Vdc; Ec1/Ib = 250 mAdc (see notes 4 and 5)	---	---	T (anode core)	---	250	°C
---	Pressure drop	7580W	No voltages (see notes 4 and 6)	---	---	---	---	0.35	Inches H_2O
		8930	No voltages (see notes 4 and 6)	---	---	---	---	0.70	Inches H_2O

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life test (3)	7580W	Group D; rf useful output power, except air-flow = 1.5 cfm; t = 100 hours (see note 4)	---	---	---	---	---	---
---	Life-test (3) end points:								
2214	RF useful output power			---	---	Po	200	---	W
1011	Humidity	7580W	See note 4	---	---	---	---	---	---
---	Humidity end point		Total grid current	---	---	Ic1	---	-15	µAdc
1032	Vibration, mechanical	7580W	Ef = 6.0V; Ebb = 2,500 Vdc; Ec2 = 500 Vdc; Rp = 4,900 ohms; Ec1/Ib = 100 mAdc; Accel = 10 G peak (min); F = 28 to 2,000 to 28 Hz (see notes 4 and 12)	---	---	Ep	---	30	Vac
---	Vibration, mechanical end points:								
1266	Total grid current			---	---	Ic1	---	-20	µAdc
1261	Electrode voltage (grid)			---	---	Ec1	-70	-100	Vdc
1042	Shock, specified pulse	7580W	Eb = 2,000 Vdc; Ec2 = 500 Vdc; Ec1 = -200 Vdc; shock = 11 ±2 ms; accel = 90 G peak (min); total impacts = 18 (see notes 4 and 13)	---	---	---	---	---	---
---	Shock, specified pulse end points:								
1266	Total grid current			---	---	Ic1	---	-20	µAdc
1261	Electrode voltage (grid)			---	---	Ec1	-70	-100	Vdc
2204	Linear amplifier power output and distortion	7580W	Eb = 2,000 Vdc; Ec2 = 400 Vdc Ec1/Ibo = 75 mAdc; Eg1/Ib = 250 mAdc 1-tone; RL = 4,000 ±100 ohms; Rg = 1,000 ohms(max); F = 2 MHz (min), 10 MHz (max); anode tank Q = 10 to 15; t = 180 (min) (see note 4)	---	---	Po 3rd IM 5th IM	250 -22 -27	--- --- ---	W(use-ful) dB dB

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NOTES:

1. To obtain maximum life, it is necessary to adjust heater voltage to values indicated below at the indicated frequencies of operation. These figures are for straight-through amplifier operation. In no case shall the heater be operated at less than 5.4 volts.

Frequency (MHz)	Ef (Vac)
300 or lower	6.00
301 to 400	5.75
401 to 500	5.50

It is recommended that the heater voltage be maintained within ±5 percent when consistent operation and extended tube life are factors. This applies to both nominal and derated voltages.

2. At the specified anode dissipation and an incoming air temperature of 25°C maximum, a minimum airflow, as specified, shall pass through the anode cooler. At this flow-rate, the static pressure drop across the tube and socket, with an appropriate chimney (air director) around the anode cooler, is approximately as specified below. The pressure drop varies with the amount of escaping air and with the shape and construction of the air director. This rating applies at bias voltages less than 100 volts and frequencies less than 500 MHz. Air cooling on the tube base shall be increased with increasing negative grid bias or with increasing frequency, or a combination of both. In all cases of operation, a socket which provides forced-air cooling shall be used and maximum seal and anode core temperature ratings shall not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them. Where emphasis is placed on long and reliable life, cooling in excess of minimum requirements shall be used.

Type	Anode dissipation (watts)	Airflow (cfm)	Static pressure drop (inch of water at sea level)	Socket
7580W	250	3.8	0.30	246-JAN
8930	350	6.0	0.60	246-JAN

3. In all electrical tests involving application of heater voltage, the use of an air-system socket and forced-air cooling is permissible.

(G) 4. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

5. The cooling test shall be made as follows: At an ambient temperature of 25°C, both base and anode shall be cooled by applying an airflow of; 7580W, 3.8 cfm maximum; 8930, 6.0 cfm maximum; at sea level from a single source using the infinite baffle system as shown on figure 2, or equivalent. At the specified test conditions, the anode core temperature shall not exceed the specified limit. Temperature shall be measured by means of a thermocouple, located as follows: The thermocouple shall be embedded in the top of the anode core, adjacent to the cooler, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole so as to hold the thermocouple firmly in place. Good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated before the cooling test can be performed.

6. An infinite baffle system as shown on figure 2, or equivalent, shall be used. Airflow shall be as follows: Type 7580W, 3.8 cfm at sea level; Type 8930, 6.0 cfm at sea level. The static pressure drop is measured across the tube and socket, with an air director (chimney) in place, to direct the total airflow through the anode cooler assembly of the tube under test.

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NOTES: - Continued

7. During the performance of life test (2), the rf grid driving voltage and the bias voltage shall be monitored and held constant. The bias voltage shall be adjusted to produce the specified value of Ibo and the rf drive shall be adjusted to produce the specified anode current of 250 mAdc during an initial adjustment period not to exceed 4 hours. In no case shall the grid be driven positive with respect to the cathode as indicated by a grid current of 50 µAdc maximum. At the conclusion of the life-test period, the change in anode current from the initial value (ΔIb), and the power output (Po) shall be noted and the tube shall meet the listed requirements for these parameters. Prior to the performance of life test (2), the tube shall have met the requirements of the linear amplifier power output and distortion test. A record shall be kept of the bias voltage used to produce the specified Ibo and the amplitude of the two-tone driving signal used in the distortion test initially, and when life test (2) is completed, the 3rd and 5th IM distortion shall again be measured, using this same value of bias voltage (irrespective of the value of Ibo this voltage may produce) and this same value of drive signal, and the tube shall meet the listed end-point requirements.

8. No other voltages shall be applied except for the test value shown. The test voltage, with the indicated polarity, shall be held for 30 seconds minimum, with no apparent arcing and a leakage current not to exceed the limit shown (see figure 4).

9. Tube type 8930 is produced by using a larger diameter cooler on the basic 7580W tube which has met all the requirements listed herein. After replacement with the larger cooler, the 8930 tube shall meet the additional requirements as specified herein.

10. The leakage current reading shall be taken during the 3 minutes immediately following the required 2-minute warmup period.

11. Circuit and cavity shall be in accordance with Drawing 223-JAN.

12. Each tube under test shall be subjected to one sweep cycle in each of the three axes X, Y, and Z. The X axis shall be defined as normal to a plane drawn through base pins No. 1 and 5. The Y axis shall be defined as normal to a plane drawn through base pins No. 3 and 7. One sweep cycle (28 to 2,000 and return to 28 Hz) shall be covered in 6 to 12 minutes. The specified voltages shall be applied during the test using the basic test circuit shown on figure 3. Tubes found to electrically oscillate for causes other than vibration shall not be tested nor rejected on this test. Each tube shall be vibrated for 60 seconds at the frequency which gives the maximum vibration (noise) output voltage in each of the three axes. If at the end of the 60 seconds the vibration output voltage is increasing, the vibration shall be continued until there is no further increase. The tubes shall not show noise-voltage output in excess of the maximum limit specified, except one intermittent short per tube shall be allowable during this test. In addition to reading noise voltage output on the HP400D, or equivalent, a permanent recording shall be made using a good-quality recorder to produce a plot of noise voltage versus frequency. Noise-voltage amplifiers used with the recorder shall have a ±1 dB frequency response over the range to be measured and the overall recording equipment shall be capable of fast response in order to show sharp noise voltage spikes resulting from internal tube resonances or other phenomena. Prominent noise peaks indicated on the recording shall be individually investigated by fixed-frequency operation, and the 60-second operation shall be made at the frequency of highest noise as so selected. Frequency at the extremes of the sweep shall be read with an accuracy of ±1 Hz below 100 Hz and 1 percent above 100 Hz.

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NOTES: -Continued

13. The tube shall be subjected to the specified acceleration in the X, Y, and Z axes, with six shocks in each axis. Tubes showing any permanent shorts, or more than one temporary short during the test, shall be rejected. The applied shock shall be an approximate half-sine wave motion with duration measured at the zero axis level.

14. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

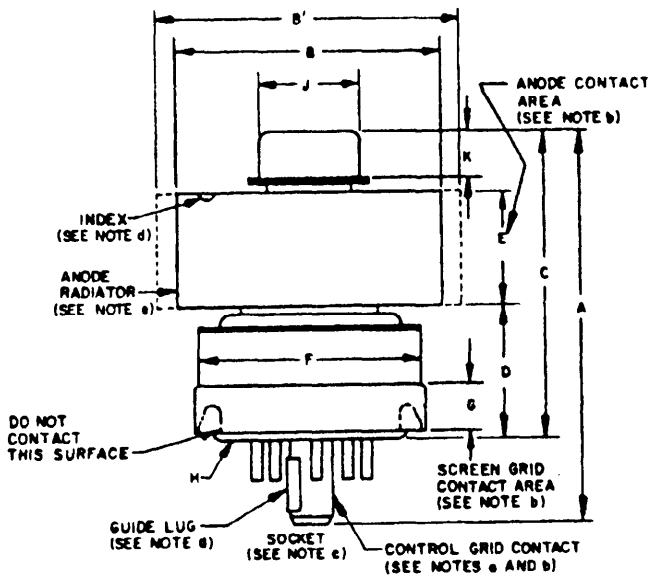
User activities:
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Preparing activity:
Navy - EC

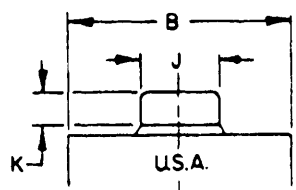
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PIN CONNECTIONS	
Pin No.	Element
1	g^2
2	k
3	h
4	k
5	int con
6	k
7	h
8	k
Center post	g^1
Radiator	a
Base ring	g^2



ALTERNATE CAP OUTLINE DRAWING

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Tube Type
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	2.324 (59.03)	2.464 (62.59)	Both
C	1.810 (45.97)	1.910 (48.51)	Both
(G) Quality conformance inspection, part 3 (see note f)			
B	1.610 (40.89)	1.640 (41.66)	7580W
B'	2.050 (52.07)	2.080 (52.83)	8930
D	.750 (19.05)	.810 (20.57)	Both
E	.710 (18.03)	.790 (20.07)	Both
F		1.406 (35.71)	Both
G	.187 (4.75)		Both
H	Base: B8: 236 (EIA) (see note a)		Both
J	.559 (14.20)	.573 (14.55)	Both
K	.240 (6.10)		Both

NOTES:

a. Pin alinement shall be checked by means of JEDEC gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gage specified on Drawing 246-JAN and shall be quality conformance inspection, part 2.

b. Alinement of anode, screen-grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality conformance inspection, part 2, shall apply. (Applies to tube type 7580W).

c. Air system socket shall be as specified on Drawing 246-JAN.

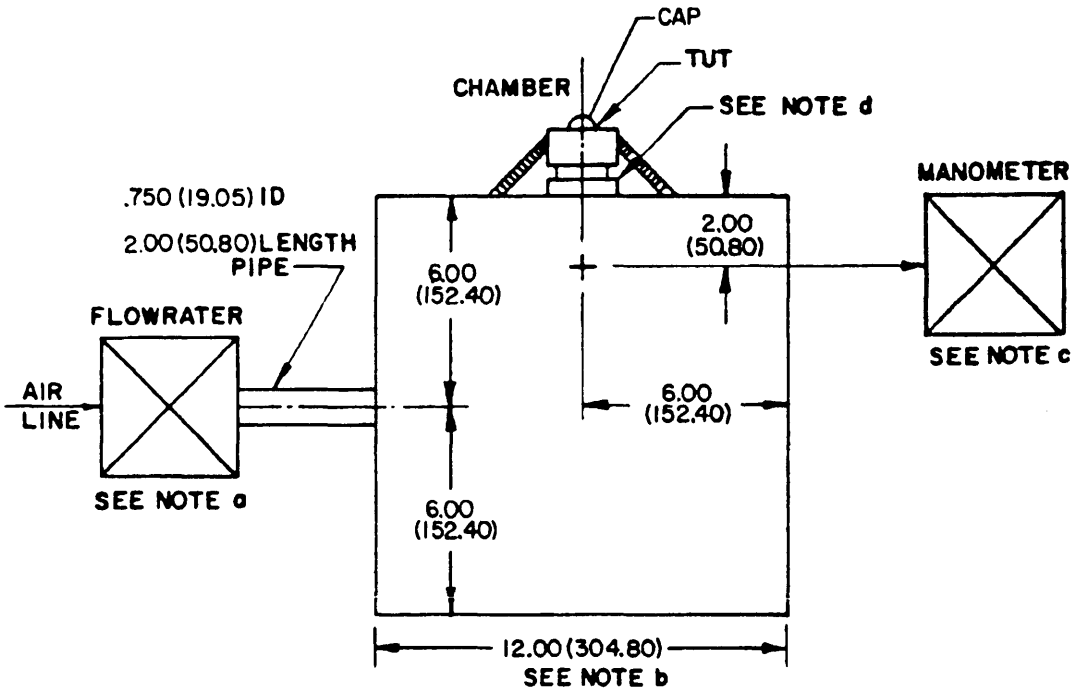
d. Location of guide lug of control-grid contact shall be referenced by a notch or arrow on the anode radiator in position shown.

e. Anode clamping shall be confined to anode radiator.

f. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube types 7580W and 8930.

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NOTES:
a. Fischer Porter Flowrater Model B4-27-10/77, or equivalent.
b. Twelve-inch cube inside dimensions, compound sealed.
c. F. W. Dwyer Manometer, 0 to 1 inch of water (Fischer Scientific Company 11-295-5 draft gage, or equivalent).
d. Socket shown on Drawing 246-JAN.
e. Dimensions are in inches.
f. Metric equivalents are in parentheses.

FIGURE 2. Block diagram.

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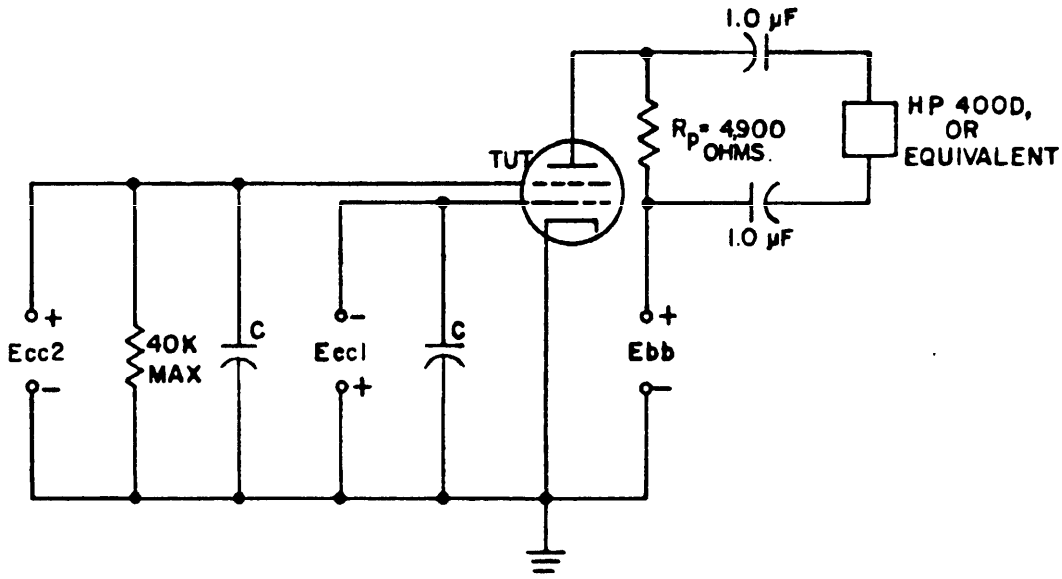


FIGURE 3. Basic test circuit for vibration test (noise).

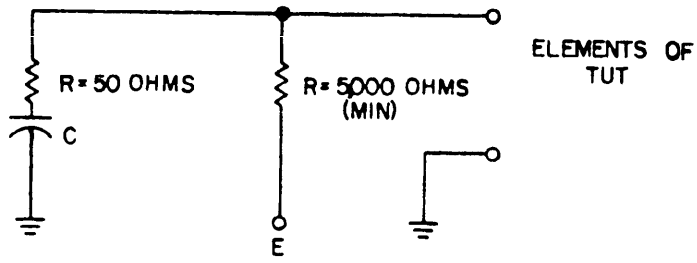


FIGURE 4. Basic test circuit for interelement leakage test.

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